AIGR TreeGrafter Stress Test — PP_1969 (Q88LG4), GO:0061798 GTP 3',8'-cyclase

Executive Judgment

Verdict: SUPPORTED. PP_1969 (UniProt Q88LG4, *Pseudomonas putida* KT2440) is a bona fide **MoaA** ortholog, and GO:0061798 (**GTP 3',8'-cyclase activity**) is the correct, appropriately *specific* molecular function. None of the three characteristic TreeGrafter failure modes apply:

1. **Granularity — PASS.** GO:0061798 is the subfamily-specific MoaA term (not a broad family label). The protein maps to PANTHER **PTHR22960:SF0** ("molybdenum cofactor biosynthesis protein 1" = MoaA/MOCS1A).
2. **Pseudo-enzyme — PASS.** All six catalytic cysteines of **both** [4Fe-4S] clusters are conserved and correctly spaced. The active site is intact.
3. **Within-superfamily mis-placement — PASS.** The diagnostic MoaA C-terminal **"twitch" domain** (Pfam PF06463 / InterPro IPR010505) is present, placing it in the true MoaA subfamily rather than a neighboring radical-SAM activity.

Single most decisive evidence: 1:1 conservation of the C-terminal GTP-binding [4Fe-4S] cluster II ligand cysteines (PP_1969 Cys268/Cys271/Cys285 ↔ E. coli MoaA Cys257/Cys260/Cys274) within the conserved twitch motif — this cluster directly anchors the GTP substrate and defines GTP 3',8'-cyclase catalysis.

Paralogy caveat (resolved, Iteration 2): *P. putida* KT2440 carries **three** MoaA-type radical-SAM paralogs — Q88E69 (the gene formally named *moaA*), Q88K11, and Q88LG4/PP_1969. This is a textbook failure-mode-1 trap, but it does **not** overturn the call: PP_1969 is actually the paralog **most** similar to the characterized *E. coli* MoaA (40.9% vs 37.5% for the named *moaA*), and all three paralogs retain the full six-cysteine active site. PP_1969's assignment is therefore well justified; the multiplicity reflects lineage-specific MoaA expansion, not functional reassignment.

Independent Family/Function Assignment

- **Most likely specific function:** GTP 3',8'-cyclase (MoaA), first step of molybdopterin (Moco) biosynthesis: 5'-GTP → 3',8-cyclo-7,8-dihydro-GTP (3',8-cH$_2$GTP), a radical-SAM reaction using two [4Fe-4S] clusters.
- **Characterized homolog basis:** *E. coli* MoaA (P30745), 40.0% identity over 315 aligned residues; mechanism/structure from *S. aureus*/E. coli MoaA (Hänzelmann & Schindelin).
- **GO term I would assign:** GO:0061798 (GTP 3',8'-cyclase activity). **Granularity vs seed: SAME** (correct and specific).
- Supporting terms consistent with assignment: GO:0051539 (4Fe-4S binding), GO:0006777 (Mo-molybdopterin cofactor biosynthetic process).

Evidence Matrix

Citation	Evidence type	Call	Claim tested	Key finding	Context	Confiden limits
P 16632608 (Hänzelmann & Schindelin 2006)	Structural/ biochemical (direct)	Supports	MoaA = GTP→precursor-Z enzyme using two [4Fe-4S] clusters	MoaA binds 5'-GTP at C-terminal [4Fe-4S] cluster; N-terminal cluster cleaves SAM	*E. coli*/*S. aureus* MoaA, crystal + biochem	High; reference enzyme, PSEPK directly
P 41183211 (Pang et al. 2025)	Mechanistic (direct)	Supports	MoaA is a radical SAM GTP 3',8-cyclase	Confirms current term/ mechanism nomenclature	Radical SAM GTP 3',8-cyclase	High
UniProt Q88LG4 xrefs	Review/ database + computational	Supports	Domain architecture = MoaA	PF04055 + PF06463; IPR040064 MoaA-like; IPR010505 twitch; PANTHER SF0	PSEPK PP_1969	High
This report: NW alignment vs P30745	Computational (evolutionary)	Supports	Ortholog with intact active site	40.9% id; 6/6 cluster cysteines conserved & spaced	PP_1969 vs E. coli MoaA	High; heuristic aligner, b motifs unambigu
This report: paralog scan (UniProt, Iter 2)	Computational (evolutionary)	Qualifies/ supports	Is PP_1969 the MoaA or a diverged paralog?	3 MoaA paralogs in KT2440; PP_1969 is CLOSEST to E. coli MoaA; all 3 keep 6/6 cys	PSEPK	High
This report: AlphaFold AF-	Computational (structural)	Supports	Are both [4Fe-4S] sites	SG–SG 5.8–6.9 Å per triad; pLDDT 86–95;	PP_1969 model	High for geometry predicted

Citation	Evidence type	Call	Claim tested	Key finding	Context	Confiden limits
Q88LG4 v6 (Iter 2)			geometrically intact?	21.4 Å inter-cluster cleft		metals in model)
UniProt Q88LG4 GO:0061799 (IEA:TreeGrafter)	Database propagation	Competing/ qualifies	Does PP_1969 also do the MoaC step?	No MoaC (PF01967) domain present → sibling-term error	PSEPK PP_1969	High

Active-Site / Placement Analysis

Pairwise alignment PP_1969 (Q88LG4) vs characterized E. coli MoaA (P30745): 40.0% identity (126/315 aligned positions).

Cluster	Role	E. coli MoaA	PP_1969	Status
N-term radical-SAM, Cx_3Cx_2C	SAM reductive cleavage → 5'-dAdo•	Cys24 / Cys28 / Cys31	**Cys36** / **Cys40** / **Cys43**	Conserved, correct spacing
C-term cluster II (GTP-binding)	Anchors 5'-GTP substrate	Cys257 / Cys260 / Cys274	**Cys268** / **Cys271** / **Cys285**	Conserved, correct spacing

C-terminal motif alignment (twitch domain):

```
E.coli P30745:  D F C A T C N R L R V S S I G K L H L C L   (257,260,...,274)
PP_1969 Q88LG4: P F C R T C S R L R L S S T G W L H G C L   (268,271,...,285)
```

Conserved GGEP motif present (…ITGGEPLV…, ~res 86). **All catalytic/metal-binding residues are intact** — no evidence of pseudo-enzyme degeneracy.

3D geometry (AlphaFold AF-Q88LG4-F1 v6, mean pLDDT 90):

Cluster	Cys triad	SG–SG distances (Å)	pLDDT
N-term radical-SAM	36 / 40 / 43	5.9, 6.9, 6.0	86–95
C-term GTP-binding	268 / 271 / 285	6.4, 5.8, 6.3	88–92

Both triads show SG–SG spacing consistent with 3-Cys coordination of a [4Fe-4S] cluster (one unique Fe left open for SAM cleavage / GTP binding respectively). The two cluster centroids sit 21.4 Å apart, flanking the central substrate cleft — the canonical MoaA arrangement. (Metals are not present in the AF model; geometry is inferred from the apo cysteine positions.)

Paralog placement (UniProt PANTHER PTHR22960, taxon 160488):

Accession	Gene	Length	% id to E. coli MoaA	6-Cys active site
Q88LG4 (PP_1969, target)	—	337	**40.9%**	6/6 intact
Q88E69	*moaA*	334	37.5%	6/6 intact
Q88K11	—	322	36.3%	6/6 intact

PP_1969 and Q88K11 are a recent duplication (67.0% mutual identity). PP_1969 is the **closest** paralog to the characterized cyclase → not a diverged outlier.

Twitch / GTP-binding region conservation (Iteration 3). Aligning the C-terminal substrate-binding region (E. coli cols ~248–282) shows the cluster-II motif intact in all three paralogs:

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E.coli : ...DFCATCNRLRVSSIGKLHLCL...
Q88E69 : ...NFCATCNRVRLTVEGRLLLCL...
Q88K11 : ...PFCRSCSRLRLSSTGRLYGCL...
PP_1969: ...PFCRTCSRLRLSSTGWLHGCL...
```

PP_1969 vs E. coli identity is balanced across domains — N-terminal radical-SAM 41%, C-terminal twitch 39% — i.e. the GTP-recognition domain is **not** selectively eroded. This rules out the subtler form of failure-mode 1 (clusters kept but substrate specificity re-tuned).

GO Curation Implications

- **GO:0061798 (GTP 3',8'-cyclase activity): RETAIN.** The IEA TreeGrafter call is correct and at the right granularity; it can be upgraded in confidence given strong orthology + full active-site conservation (lead for curator verification).

- **GO:0061799 (cyclic pyranopterin monophosphate synthase activity): REMOVE / replace.** This is the MoaC activity; PP_1969 has no MoaC domain. It is a sibling-activity carry-over from the mixed PANTHER family and should not sit on a MoaA protein (curator verification recommended).
- Keep GO:0051539 (4Fe-4S binding) and GO:0006777 (Moco biosynthetic process).

Conflicts, Knowledge Gaps, Discriminating Tests

- **Paralog check (done):** *P. putida* KT2440 has **three** MoaA paralogs (Q88E69=*moaA*, Q88K11, Q88LG4=PP_1969). All retain the full 6-Cys active site, and PP_1969 is the closest to E. coli MoaA. This redundancy means the *in vivo* roles may be partitioned (e.g., condition-specific expression, aerobic vs anaerobic Moco demand, or distinct molybdoenzyme clients). The MF term is justified for PP_1969, but which paralog(s) carry flux in a given condition is an open experimental question and a candidate for transcriptomic/knockout study.
- **No direct PSEPK assay:** The function is inferred from orthology + residue conservation, not from a *P. putida*-specific enzymatic assay. Confidence is high but remains computational/evolutionary for this exact organism.
- **Most efficient discriminating experiment:** heterologous expression + in vitro radical-SAM assay measuring 5'-GTP → 3',8-cH_2GTP conversion (anaerobic, with dithionite/SAM), or complementation of an *E. coli ΔmoaA* mutant for Moco-/nitrate reductase–dependent growth. Either would directly confirm GTP 3',8'-cyclase activity.
- **Database carry-over caveat:** the GO:0061799 co-annotation illustrates that a single PANTHER family spanning MoaA and MoaC can emit both terms; curators should separate them by domain architecture.

Conclusion

PP_1969 directly has the stated function. GO:0061798 is **supported** and specific; the only curation fix is removing the co-propagated MoaC sibling term GO:0061799.